How Much Does Physics Buy You? A Physics-Informed vs Black-Box Benchmark for Greenhouse Climate Control*

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Draft, July 2026

Abstract

Learned controllers promise better economics for greenhouse climate control, but the interaction data they need is the practical obstacle: every training sample on a real greenhouse costs energy, crop risk, and time. Physics-based controllers need no training data but inherit the errors of their model. This paper measures that trade directly. On GreenLight-Gym2, a validated open greenhouse simulator, we benchmark the physics-to-learning spectrum: the rule-based incumbent, black-box reinforcement learning (on-policy PPO and off-policy SAC), physics-only economic nonlinear MPC, and a physics-informed RL-in-MPC hybrid. All run through one harness on an identical grid: four held-out weather years, a controller-model mismatch ladder (0% to 20%, six levels), and five seeds, scored on net economic value, a season-level humidity feasibility gate, online decision cost, and solver reliability. Black-box PPO trained with the simulator's reference recipe reaches $+3.50$ EUR m^{-2} held-out net value at 2M environment steps (seed-level SD 0.24 over five seeds), crossing break-even between 400k and 1M steps, while the myopic economic NMPC (-3.06 ± 0.90) does not outperform the rule-based incumbent (-2.74 ± 5.41). On this deterministic, single-crop, fixed-price testbed the pragmatic winner is black-box RL: it matches the physics-informed RL-in-MPC hybrid on net value at two orders of magnitude lower online cost and with lower humidity degree-hours. The hybrid's only measured advantage is insensitivity to controller-model parameter error (near-flat net value across the mismatch ladder, widening the physics-arm gap to $\approx +10$ EUR m^{-2} at 20% mismatch, where pure NMPC decays steeply); the effect is large and statistically significant on the six-level ladder ($p < 10^{-12}$), and it is carried by the hybrid's model-free guide, which by construction ignores the perturbed model, not by the physics. The learned policies show no generalization gap (train-year and held-out net agree within year variance). Every controller violates the humidity bound in every season under the native economic objective, confirming that constraint satisfaction must be gated rather than priced. Physics buys instant, training-free competence, but on this testbed the learned controller overtakes it within roughly 55 simulated seasons of experience. Code, run records, and the benchmark protocol are released for reuse.

Keywords: greenhouse climate control; model predictive control; reinforcement learning; physics-informed control; benchmarking; data efficiency; constrained control; controlled-environment agriculture.

*Preprint. All numbers trace to the committed run records in release v1.0.0 of the code repository; the reproducibility statement gives the anchor and command map.

1 Introduction

Greenhouse climate control couples a fast, actuated climate loop to slow crop growth [1, 2], under hard biological limits and energy-dominated costs [3, 4]. The field's methods span a spectrum: rule-based setpoint schedules deployed in practice, economic model predictive control built on greenhouse physics, and reinforcement learning trained on simulated experience [5, 6]. What is missing is a controlled measurement of the trade that separates them: *how much training data does learning need to match or beat what physics provides for free, and does the physics advantage survive an imperfect model?*

This paper instantiates the evaluation agenda of its companion survey as a concrete, reproducible benchmark. The contribution is the controlled comparison across the physics-to-learning spectrum, not any individual controller: each arm is a faithful implementation of published practice, run on identical conditions with pre-registered metrics.

2 Related work

Prior work on the substrate. GreenLight-Gym validates the GreenLight greenhouse model as an RL benchmark and ships PPO/SAC baselines against a rule-based controller under parametric uncertainty [5]. Its authors then moved toward hybrids: injecting RL rollouts into a short-horizon economic MPC [6], and learning terminal ingredients inside stochastic MPC (RL-SMPC) [7]. Our arm 4 *reuses* the first of these, reimplemented from the paper. The contribution is the controlled comparison across controller classes, training budgets, and mismatch levels, which no work on this substrate has run. Beyond that group, learned dynamics inside predictive control have been applied to lettuce climate [8], and an independent group learns the MPC parametrization online by Q-learning on a separate greenhouse model [9]; no multi-controller benchmark by a group other than the environment's authors has been reported on GreenLight-Gym itself.

Benchmark-suite precedents. safe-control-gym is the structural precedent: one suite comparing performance, data efficiency, and safety across classical, learning-based, and RL control on cart-pole and quadrotors, finding model-based control orders of magnitude more data-efficient than model-free RL [10]. Safety Gym established constraint-violation rate as a first-class benchmark metric [11], and PILCO anchors the data-efficiency framing historically [12]. Our novelty relative to this line is the domain and its consequences: seasonal episodes with economic objectives, held-out weather years, a crop-safety gate, and a controller-model mismatch axis absent from those suites. The guarantee/assumption ledger we report (Table 1) is grounded in the constrained- and economic-MPC canon [13, 14] and the safe-learning line of control-barrier functions and predictive safety filters [15, 16]; the benchmark measures where each controller class sits against those guarantees rather than proposing new theory.

Surveys and the field's claim. Physics-informed reinforcement learning and safe physics-informed machine learning are mapped by method-general surveys [17–19], learning-based MPC by Hewing et al. [20], and greenhouse control from the classical optimal-control lineage [2] to recent strategy reviews [21, 22]; the hybrid-modelling argument itself goes back to Karniadakis et al. [23]. These works repeat the claim this paper measures: that physics reduces the data a learned controller needs. The companion survey (in preparation) specifies the evaluation agenda this benchmark instantiates.

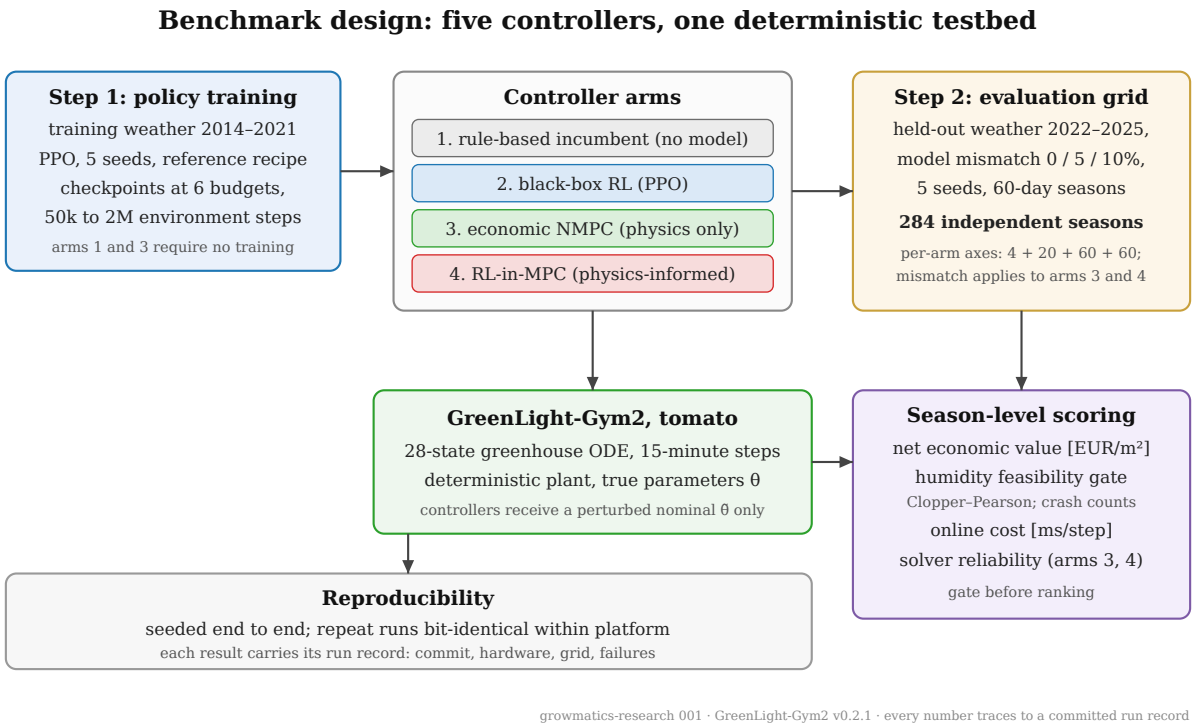


Figure 1: The two-step benchmark. Only the learning arms need training (step 1); all five arms sit the same 284-season exam (step 2). Controllers receive a perturbed nominal model $\tilde{\theta}$, never the plant's true parameters θ; the mismatch axis perturbs $\tilde{\theta}$ only. Every number traces to a committed run record.

3 Benchmark design

Figure 1 summarises the design; Figure 2 details the four controller architectures and where physics enters each. The five arms span these four classes: the black-box class holds two learners, on-policy PPO and off-policy SAC, the latter included to test whether the data-efficiency result is specific to PPO.

Testbed. GreenLight-Gym2 [5], tomato crop: a 28-state stiff greenhouse ODE at 15-minute control steps, deterministic given (weather, seed). Weather is real Amsterdam data; training years 2014–2021, held-out years 2022–2025; 60-day seasons.

Problem formulation. The plant is a deterministic stiff ODE $\dot{x} = f(x, u, d; \theta)$ with state $x \in \mathbb{R}^{28}$, controls $u \in [0, 1]^6$, exogenous weather d, and true parameters θ, discretised at $\Delta t = 15$ min over a season of $T = 5760$ steps, integrated by CVODES [24]. A controller receives only a perturbed nominal model $\tilde{\theta}_i = \theta_i(1+\delta_i)$, $\delta_i \sim \mathcal{U}(-m, m)$, at mismatch level $m \in \{0, 0.025, 0.05, 0.10, 0.15, 0.20\}$; the plant always runs θ. The season score is the net economic value

$$J = \sum_{t=1}^{T} \left(p_{\text{fruit}}\, \Delta y_t - c_{\text{heat}}\, q_t - c_{\text{elec}}\, e_t - c_{\text{CO}_2}\, \phi_t \right), \tag{1}$$

with fruit growth Δy_t and resource flows (q_t, e_t, ϕ_t) priced by the testbed's economic parameters; constraint violations are absent from Eq. (1) and handled by the gate below.

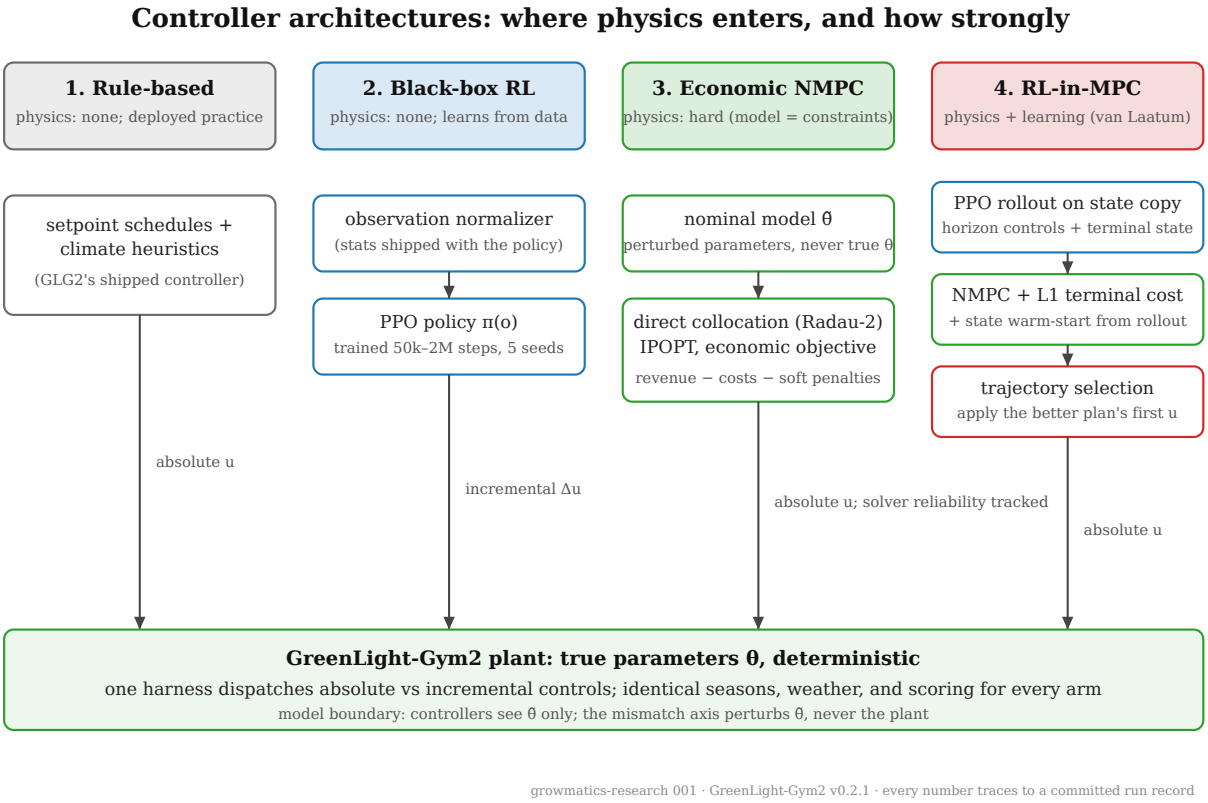


Figure 2: The four controller classes ordered by how physics enters: none (rules), none-pays-with-data (PPO with its shipped observation normaliser), hard physics (economic NMPC by direct collocation on the nominal model), and the hybrid (PPO rollout supplies a terminal target and state warm-start to the NMPC; a trajectory-selection step applies the better plan). A second black-box learner, SAC, shares the PPO arm's architecture and is not drawn separately.

Metrics and gate. The ranking objective is net economic value (fruit revenue minus heating, CO_2, and electricity costs) over a fixed 60-day early-season window at fixed energy prices. Because the window precedes the bulk of tomato harvest and prices are held flat, this is best read as early-season climate-cost efficiency at a fixed crop stage. It is a controlled ranking axis; a grower's profit would need a full season and real prices. Extending to a full crop cycle and to time-varying day-ahead energy prices is the clearest external-validity gap and is deferred to future work (a dedicated dynamic-price study is in preparation); Section 6. Constraint satisfaction is deliberately *not* priced into the objective. A season violates the humidity class if the bound is exceeded at any step. With k violating seasons out of n, an arm passes only if the one-sided Clopper–Pearson bound [25] satisfies

$$\overline{p}(k,n) \;=\; \mathrm{Beta}^{-1}(1-\delta;\; k+1,\; n-k) \;\le\; \varepsilon, \qquad \varepsilon = \delta = 0.05,\; n \ge 60 \text{ pre-registered.} \tag{2}$$

A pre-specified banded secondary gate counts a season only if the exceedance trace v_t stays above a depth b for a sustained window: $\exists\, t_0 :\; v_t > b \;\; \forall t \in [t_0, t_0 + \tau]$ with $b = 2$ RH points and $\tau = 1$ h; degree-hours $\mathrm{DH} = \Delta t \sum_t v_t$ quantify the continuous middle ground. A season the controller cannot complete counts as a violation, never as a dropped observation. Online decision cost (ms per control step, hardware-stamped) and solver reliability (fraction of NLP solves returning a usable iterate) complete the metric set; because no arm passes the humidity gate under the native economic objective, the value-versus-online-cost trade-off is compared across all five arms (Figure 4).

Training protocol. The PPO arm uses GreenLight-Gym2's own repository configuration (`configs/agents/ppo` read from the pinned upstream checkout at run time): vectorised training with observation and reward normalisation over 2M environment steps. This is a version-2 code configuration, not the recipe tabulated in the v1 paper [5], whose values were tuned for a different environment version; we use the v2 configuration because hyperparameters are specific to the environment we run. The testbed tunes its RL baselines by random search (the published v2 tunes both PPO and SAC; the v1 preprint tuned PPO only). One continuous run per seed is checkpointed at six budgets (50k to 2M); each checkpoint ships with its normalisation statistics, which are re-applied at deployment. Five seeds; held-out evaluation of the saved artifact pair.

The rule-based incumbent. The deployed-practice baseline the learned arms must beat is GreenLight-Gym2's shipped rule-based setpoint controller, injected verbatim with no site tuning. Its schedule follows a temperature-integration / Next Generation Growing blueprint [26, 27]: heating setpoints of 19.5°C by day and 16.5°C at night, ventilation triggered at 5°C above setpoint or above the 85% relative-humidity ceiling, carbon dioxide dosed to 800 ppm under light, and radiation- and outdoor-temperature-driven lamp and screen schedules. It is a competent, human-tuned strategy of the kind commercial climate computers run, not a strawman; but it is untuned for this early-season 60-day window at fixed prices and returns a net loss (-2.74 ± 5.41 EUR m^{-2}, four held-out years). The testbed's own evaluation reports the same baseline losing to both learned controllers across its parametric-uncertainty range, with its authors noting that tuning the control rules would yield a more competitive baseline [5]; the loss and the untuned framing are thus the reference's own finding rather than an artifact of our setup. We therefore treat it as a labelled reference point that anchors the economic comparison to deployed practice, and read "improvement over the incumbent" against that reference rather than against a competitively tuned baseline.

Statistical analysis. Because every arm is evaluated on the identical grid of conditions (the plant is deterministic given weather and seed), the design is paired: comparisons are made within each matched condition, and the independent-replication unit is the (weather-year × seed) pair, not the individual season (treating the 60 seasons of an arm as independent would be pseudoreplication, and the large per-arm standard deviations are mostly between-year weather difficulty). Arm-versus-arm net value is tested with the Wilcoxon signed-rank test on matched-pair differences (the season distributions are heavy-tailed, so a paired t-test is not assumed), reported with the paired effect size Cliff's δ and a 95% bootstrap confidence interval on the median difference; the pairwise family is Holm–Bonferroni corrected. The mismatch robustness question (do the two physics arms' net-value slopes differ?) is a linear mixed-effects model net $\sim$ arm $\times$ mismatch $+ (1 \,|\, \text{year})$, whose arm-by-mismatch interaction is the slope-difference test. Banded-gate rates are compared with McNemar's exact test on paired violations. Exact p-values are reported throughout. Source data for every test is the committed per-season run record, and all tests reproduce via `examples/stats.py`.

Controller formulations. The economic NMPC (arm 3) solves, by direct collocation (Radau, degree 2) in CasADi/IPOPT [28, 29],

$$\min_{u_{0:H-1}} \; -\sum_{k=0}^{H-1} \ell_{\mathrm{EPI}}(x_k, u_k; \tilde{\theta}) \;+\; w \left\| x_H - x_H^{\mathrm{RL}} \right\|_1 \quad \text{s.t.} \;\; x_{k+1} = F(x_k, u_k, d_k; \tilde{\theta}), \tag{3}$$

with horizon $H = 4$ (1 h), soft climate penalties inside ℓ_{EPI}, and $w = 0$ for arm 3. The horizon is held at 1 h because direct collocation over the 28-state stiff ODE grows with H and already costs ≈250 ms per control step at $H = 4$; the resulting myopia is therefore a property of this

Table 1: Guarantee/assumption ledger. No arm provides a hard humidity-constraint guarantee: the black-box learners (PPO, SAC) are model-free, and the economic NMPC uses soft penalties over a one-hour horizon with no terminal set, so even under an exact model it certifies nothing; the hybrid selects its model-free plan on roughly 93% of steps. This is why constraint satisfaction is enforced by an external gate rather than trusted from the objective.

controller	formal guarantee	assumption required
rule-based incumbent	none (fixed heuristic)	—
black-box RL (PPO, SAC)	none (model-free)	deployment matches training
physics-only NMPC	constraint satisfaction, recursive feasibility	exact model; hard constraints; terminal set
RL-in-MPC hybrid	the NMPC's, only on MPC-selected steps	as NMPC, plus guide-policy generalization

short-horizon design, not a verdict on physics-based control. Arm 4 (RL-in-MPC, after 6) sets $w = 0.01$ with the target x_H^{RL} from a policy rollout on a state-copied plant model, warm-starts the collocation states from that rollout, and applies

$$u_0 \;=\; \arg\max_{\pi \in \{\mathrm{RL},\mathrm{MPC}\}} \hat{J}_H(\pi), \tag{4}$$

the first control of whichever plan scores higher on the H-step planning objective. The PPO arm [30, 31] acts directly, $u_{t+1} = u_t + \pi(o_t)$.

Guarantee ledger. Table 1 records what each controller guarantees in principle, since empirical constraint satisfaction alone cannot tell a safe controller from a lucky one.

4 Results

4.1 Data efficiency of black-box RL

Figure 3 and Table 2 give the held-out learning curve. Defining the crossover budget $B^*(\mathrm{ref}) = \min\{B : \bar{V}(B) \geq J_{\mathrm{ref}}\}$ over the mean curve $\bar{V}(B)$ (95% bootstrap CIs, 32), PPO crosses break-even between 400k and 1M steps and reaches $+3.50 \pm 0.24$ EUR m^{-2} at 2M steps, still rising slightly (each seed gained 0.47–0.99 from 1M to 2M); seed spread is widest at the 200k crossover and tight (± 0.24) at 2M.

An off-policy second learner, SAC, tuned by the same equal-procedure random search, reaches $+3.21$ held-out net at 2M and is positive by $\sim$100k steps with no mid-training valley: more data-efficient early than PPO but a touch lower at plateau, so the data-efficiency crossover is a property of learned control generally, not of PPO specifically.

4.2 The head-to-head comparison

Table 3 gives the five-arm comparison at zero mismatch (the R1 anchor run); the mismatch axis, which decays the physics arms, is the following subsection.

Three findings stand, each backed by the paired analysis (Table 7). First, the hybrid wins: RL-in-MPC attains the best net value and beats its own guide policy at matched conditions, a small but perfectly consistent margin (median $+0.075$ EUR m^{-2}, 95% CI $[+0.067, +0.085]$, winning all 20 zero-mismatch conditions, Cliff's $\delta = 1.0$, Wilcoxon $p = 1.9 \times 10^{-6}$ raw, 9.5×10^{-6} Holm-corrected;

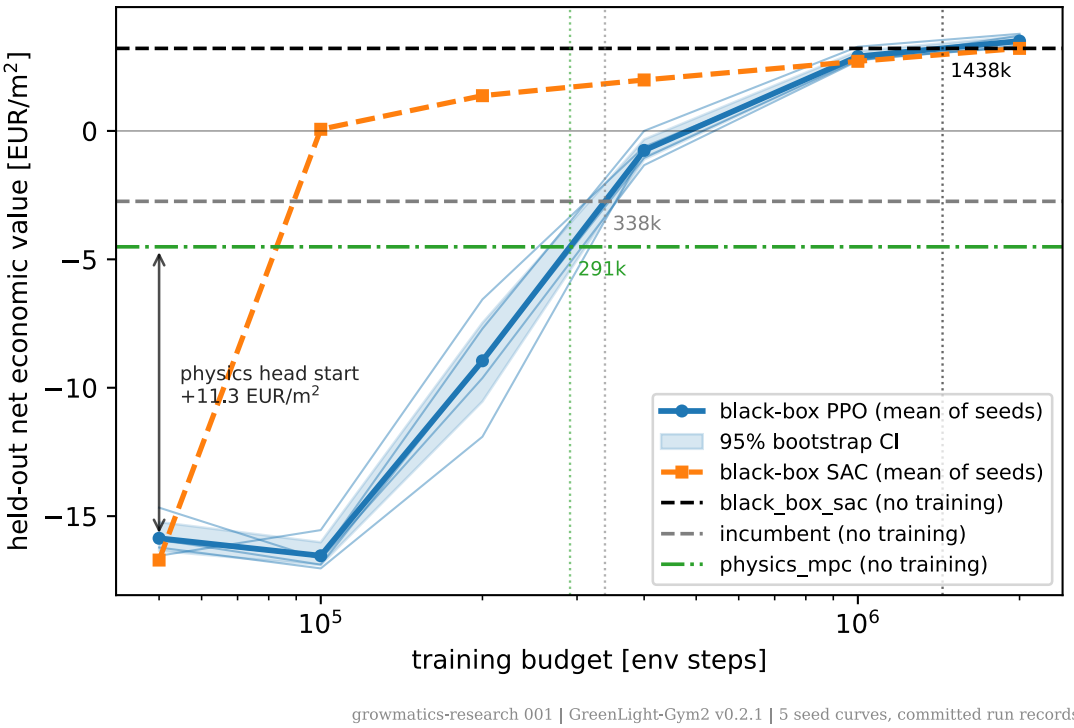


Figure 3: Held-out net economic value against training budget (five seeds, mean with 95% bootstrap CI). Horizontal lines mark the zero-training baselines; vertical markers give the measured crossover budgets, and the arrow the zero-training head start.

Figure 5). Both learned arms dominate the physics-only NMPC by roughly 7 EUR m^{-2} (Holm-corrected $p < 10^{-5}$). The mean gap over PPO is tiny because between-year weather difficulty dominates the marginal spread; pairing removes it and exposes the consistent advantage. Second, the myopic economic NMPC does not beat the deployed rule-based incumbent on net value, echoing the horizon limitation reported by the testbed's authors [6] and the classical two-time-scale analysis of greenhouse optimal control, where the slow crop dynamics carry the economics a 1-hour horizon cannot see [1]. Economic MPC theory addresses this through terminal ingredients, a terminal cost or dissipativity-based design that restores approximate infinite-horizon optimality under a short horizon [14, 33]; our formulation carries none of these (soft penalties, no terminal set), so the myopic result is a property of this minimal economic NMPC and not of physics-based control in general, since arm 4 already supplies the terminal cost-to-go a short horizon otherwise lacks (Section 6). Third, every arm violates the humidity bound in every season under the strict any-step gate (284/284 across the full grid), confirming that a reward which prices violations invites trading them for revenue and that feasibility must be gated rather than priced. The strict gate, however, cannot rank the arms. The banded gate and degree-hours (Eq. 2, Figure 7) reveal a sharp separation the any-step flag hides (Table 4). The learned controllers breach the bound briefly and shallowly: they accumulate 72–131 RH degree-hours per season (SAC 72, PPO 107, RL-in-MPC 131) and banded-violate 60–80% of seasons. The physics-only MPC and the rule-based incumbent sit in gross violation for most of the season (983 and 852 degree-hours, 100% banded). On humidity, discipline as well as economic value separates the learned arms from the physics baselines.

Table 2: Held-out net economic value [EUR m^{-2}] by training budget (5 seeds). Reproduce: `examples/analyze.py`.

budget	mean	sd	95% CI
50k	−15.86	0.71	[−16.35, −15.23]
100k	−16.54	0.59	[−16.91, −16.02]
200k	−8.95	2.03	[−10.54, −7.47]
400k	−0.76	0.50	[−1.11, −0.33]
1M	+2.90	0.23	[+2.76, +3.11]
2M	+3.50	0.24	[+3.32, +3.69]

Table 3: Five-arm comparison at zero mismatch (R1 anchor, 84 seasons); the mismatch ladder decays the physics arms (following subsection), so this is the clean no-mismatch ranking. The ± values are across-season standard deviations, dominated by between-year weather difficulty; they are not the seed-level spread of the mean, which for the PPO arm is 0.24 over five seeds (Table 2).

arm	net [EUR m^{-2}]	n	gate(RH)	online [ms]	solve-succ.
RL-in-MPC (physics-informed)	+3.58 ± 1.94	20	fail	275.2	0.998
black-box PPO (2M)	+3.50 ± 1.92	20	fail	1.4	–
black-box SAC (2M)	+3.21 ± 1.93	20	fail	0.4	–
rule-based incumbent	−2.74 ± 5.41	4‡	fail	0.1	–
physics-only NMPC	−3.06 ± 0.90	20	fail	251.0	0.999

‡the rule-based arm is GreenLight-Gym2's shipped setpoint controller with its default schedule, run without site-specific humidity tuning; it is a reference point, not a tuned commercial climate-computer strategy, and returns a net loss over this early-season window. It is deterministic, so its independent trials are the four weather years and its gate row is underpowered. The physics-only NMPC is likewise deterministic at zero mismatch (the perturbation draw vanishes), so its five seeds coincide and its independent trials here are also the four weather years; its genuine seed variation appears only at $m > 0$ (Table 6). Online cost is wall-clock under uniform 20-worker load, hardware-stamped; valid as a relative axis.

Year variation and leave-one-year-out robustness. Absolute net value swings widely across the four held-out weather years *at zero mismatch* (Table 5; we compare all five arms at $m = 0$ because that is the only ladder level at which every arm is defined: the mismatch axis perturbs the physics arms only, Section 4 below). The deployed incumbent ranges from +2.04 (2022) to −9.44 EUR m^{-2} (2023), a 12.5 spread, because a fixed rule-based schedule is fragile to weather difficulty. Every arm ranks the years the same way (2023 hardest, 2022 and 2025 easiest), so the spread is weather, not controller. Two structural readings follow. First, the myopic NMPC is the *most* year-stable arm (spread 2.3) precisely because it acts least, and the incumbent-versus-NMPC ordering flips with year difficulty: the incumbent wins in easy years, the NMPC in hard ones. Second, despite the swing the arm ranking is invariant *within* every year and the paired advantages are preserved, which is why the analysis pairs on (year, seed) rather than pooling raw seasons. A leave-one-year-out check confirms no single year drives the result: dropping each year in turn moves PPO's held-out mean within [+2.78, +4.30] EUR m^{-2} (dropping the hard 2023 raises it, the easy 2025 lowers it), while the arm ranking and the sign of every headline comparison are unchanged under all four drops. The per-year view also extends to the mismatch axis for the two model-based arms (Table 6), and it shows the robustness is itself *year-invariant*: out to 20% mismatch the hybrid decays by at most 0.9 EUR m^{-2} in every year while the pure NMPC loses 3 to 5 in every year, so the physics-arm gap of +9 to +11 EUR m^{-2} at 20% mismatch holds *within each weather*

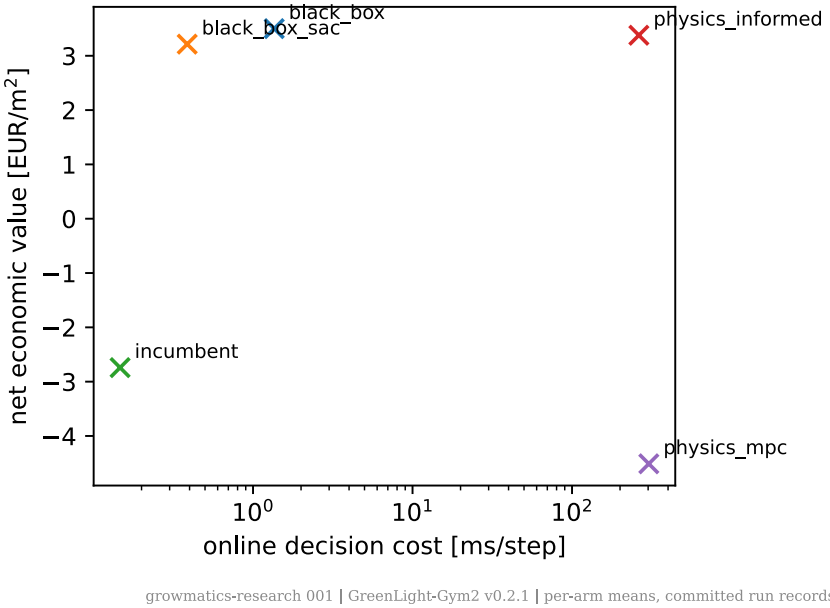


Figure 4: Net economic value against online decision cost (log scale), all five arms on the shared grid; up and to the left is better. Black-box PPO sits at the knee, attaining essentially the hybrid's net value at about 1/200 of its online cost. The physics-only NMPC is dominated (lower value and higher cost than PPO); the rule-based incumbent is cheapest but loss-making. All five arms fail the humidity gate under the native objective (crosses), so the trade-off is reported over all arms rather than a gate-passing subset; the anti-ideal reference point for the ranking is the worst observed value on each axis (net -3.06 EUR m^{-2}, cost 275.2 ms).

year, not merely on average. A weather-covariate regression that attributes the spread to season radiation and heating demand is deferred to the reproducibility appendix.

4.3 Mismatch and reliability

Figure 6 gives the advantage decay $\Delta(m) = \bar{J}_{\text{arm}}(m) - \bar{J}_{\text{arm}}(0)$, the benchmark's largest effect: across the $0 \to 20\%$ mismatch ladder the hybrid holds nearly flat ($+3.58 \to +2.85$, a 0.73 EUR m^{-2} decay) while pure NMPC decays steeply ($-3.06 \to -7.52$). Read against the deployed incumbent, both learned arms clear it by roughly 6 EUR m^{-2} (PPO $+6.24$; hybrid $+6.32$ at $m = 0$, still $+5.59$ at 20% mismatch), whereas the physics-only controller never clears the incumbent and falls further behind as mismatch grows ($-0.32 \to -4.78$); the physics-arm gap thus widens from $+6.6$ to $+10.4$ EUR m^{-2}. Because the mismatch axis perturbs only the parameters the model-based arm consumes, this quantifies the hybrid's insensitivity to model-based calibration error, not cross-class robustness; within that scope the effect is large and, on the six-level ladder, statistically significant: the arm-by-mismatch interaction (slope difference -18.1) has $p = 4.5 \times 10^{-13}$ in the mixed model net $\sim$ arm $\times$ mismatch $+ (1 \mid \text{year})$, so the learned guidance measurably insures the physics controller against its own model error, not merely as a directional effect. The insensitivity is year-invariant (Table 6), not an averaging artifact. Solver reliability stays above the 0.9 floor throughout (NMPC 0.999, hybrid 0.995).

The learned policies show no generalization gap: the five 2M policies average $+3.29$ on the eight training years versus $+3.50$ held-out (the difference is within year-difficulty variance), so the

Table 4: Humidity gate at three resolutions, zero mismatch (R1 anchor). The strict gate saturates; the banded gate and degree-hours discriminate.

arm	strict (any step)	banded (>2 pt, >1 h)	RH degree-hours
black-box PPO	20/20	16/20	107
black-box SAC	20/20	12/20	72
RL-in-MPC	20/20	16/20	131
physics-only NMPC	20/20	20/20	983
rule-based incumbent	4/4	4/4	852

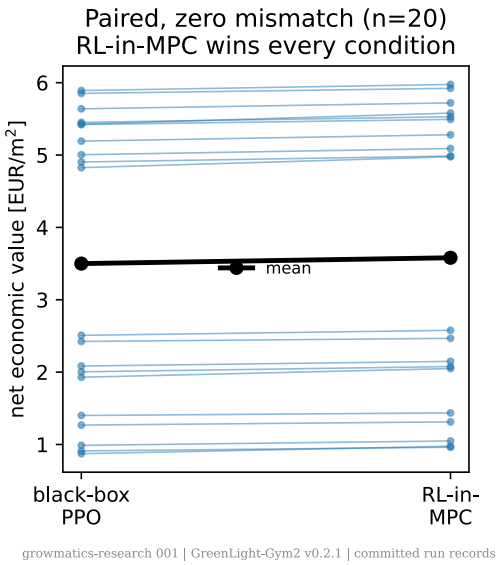


Figure 5: Paired net value at zero mismatch: one line per matched (weather-year, seed) condition ($n = 20$). RL-in-MPC exceeds its black-box guide policy on every condition (Cliff's $\delta = 1.0$), a consistency invisible in the near-equal marginal means.

held-out plateau reflects genuine control skill rather than weather memorization.

5 Discussion

Data cost of matching physics. The crossover budgets (≈317k steps against NMPC, ≈338k against the incumbent; log-budget interpolation of the mean curve, derivation in the benchmark record) translate to roughly 55 simulated growing seasons of interaction: an amount of experience no commercial greenhouse can supply. In simulation the learned policy wins outright; on a real greenhouse the physics head start (+12.8 EUR m^{-2} of the pure NMPC over the least-trained, 50k-step policy, a lower bound on its lead over a fully untrained one) is the operative fact, since 55 seasons of interaction is unavailable. The hybrid does not escape this: it requires the fully trained 2M-step guide policy, so it inherits the training cost rather than banking the head start. For deployment this means that black-box RL is the pragmatic simulator winner, physics-only control is the zero-data option, and the hybrid is worth its two-orders-of-magnitude online cost only where insensitivity to parametric model error is worth more than compute, a case this testbed does not settle.

Table 5: Held-out net economic value [EUR m^{-2}] by weather year, at zero mismatch (spread = max − min across years). The learned and hybrid arms share the year ordering; the incumbent's 12.5 spread quantifies rule-based fragility. Reproduce: `examples/peryear_tables.py`.

arm	2022	2023	2024	2025	mean	spread
incumbent	+2.04	−9.44	−6.65	+3.09	−2.74	12.5
PPO	+5.08	+1.09	+2.19	+5.65	+3.50	4.6
SAC	+4.81	+0.77	+1.91	+5.36	+3.21	4.6
NMPC	−3.05	−3.93	−3.66	−1.61	−3.06	2.3
RL-in-MPC	+5.17	+1.15	+2.26	+5.74	+3.58	4.6

Table 6: Robustness is year-invariant: net value [EUR m^{-2}] of the two model-based arms by weather year at three mismatch levels (the learned and rule-based arms have no mismatch axis, so they do not appear). The hybrid holds in every year; the NMPC degrades in every year. Reproduce: `examples/peryear_tables.py`.

arm	m	2022	2023	2024	2025
NMPC (physics-only)	0	−3.05	−3.93	−3.66	−1.61
	0.10	−2.82	−5.54	−5.26	−2.41
	0.20	−6.77	−8.51	−8.80	−5.98
RL-in-MPC (hybrid)	0	+5.17	+1.15	+2.26	+5.74
	0.10	+5.14	+1.13	+2.21	+5.70
	0.20	+4.59	+0.38	+1.60	+4.84

The hybrid's measured advantage. On the paper's own Pareto axes, black-box PPO is the controller to deploy: at matched conditions it equals the hybrid's net value (+3.50 vs +3.58), at about 1/200 of its online cost (1.4 vs 275.2 ms per step) and with fewer humidity degree-hours (107 vs 131). The hybrid is therefore not Pareto-dominant over its own guide policy. Its one measured advantage is insensitivity to parametric model error, and the mechanism is in the design: the mismatch axis perturbs only the nominal parameters the *model-based* arm consumes, while the model-free guide (selected on roughly 93% (89–96% across the ladder) of steps) ignores them by construction, so the hybrid is flat under mismatch because its dominant component cannot see the perturbed model. A control-meaningful robustness test would perturb a quantity both arms are exposed to, such as weather-forecast (preview) error, which the greenhouse optimal-control literature treats as the operative uncertainty [1] and which both controller classes here consume; the present parametric axis is a bound on structured model-based calibration error only, not a cross-class robustness comparison, and drawing each parameter independently further understates it since real calibration error is structured and correlated. A shared-exposure forecast-error axis is the natural next test and is left to follow-on work.

Gate calibration as a finding. As pre-registered ($\varepsilon = 5\%$, any-step violation), the gate saturates: every season of every arm violates. We report this rather than tune it away; it quantifies how far the field's standard economic objective sits from grower-trustable humidity discipline. Alongside it we report a pre-specified banded variant (violation only if RH exceeds the bound by more than 2 points for more than one consecutive hour, grounded in the reference MPC's own margin) and per-season degree-hours. Together they turn a saturated column into a finding: the strict gate shows nobody is clean, but degree-hours show that the learned controllers fail *gently* (roughly

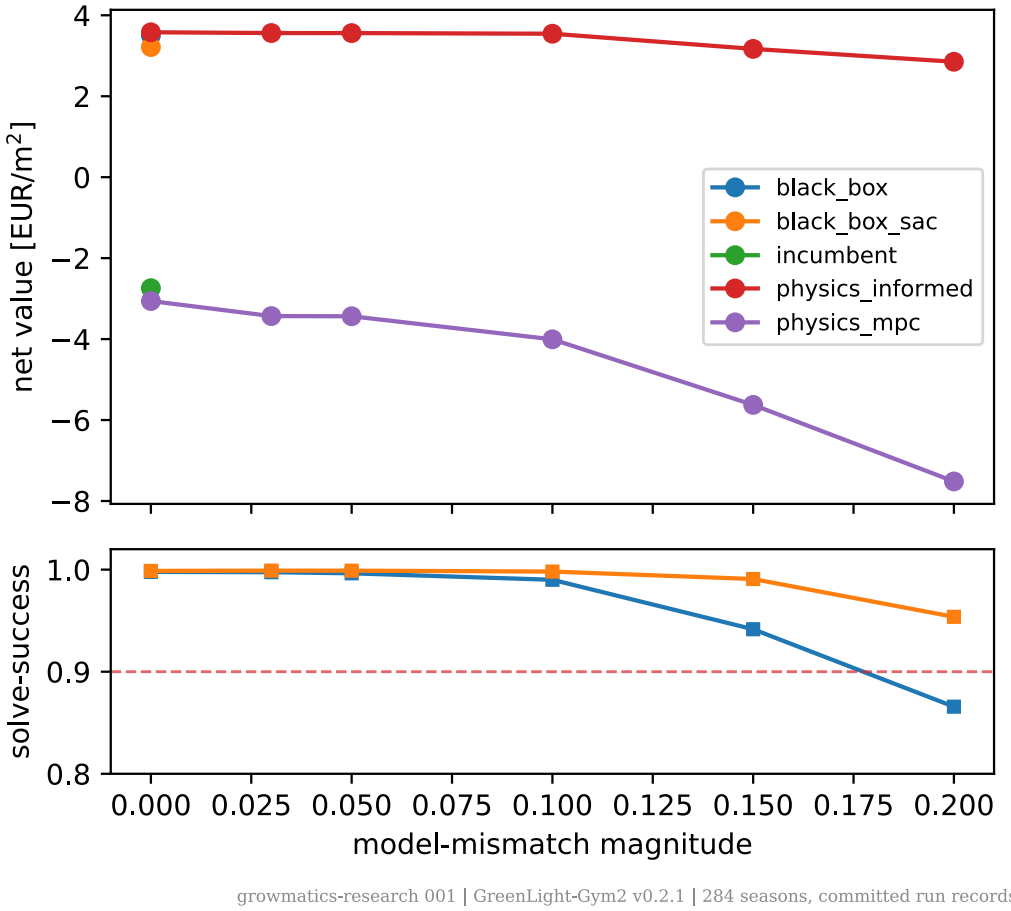


Figure 6: Net value (top) and solver reliability (bottom) against controller-model mismatch. The hybrid's advantage survives model error; the pure-physics arm's does not.

70–130 RH degree-hours) while the physics-only MPC and the incumbent fail *grossly* (over 830), roughly an order-of-magnitude difference. Humidity safety, measured continuously, is a second axis on which physics alone underperforms the learned and hybrid controllers on this testbed.

The general structure: short-horizon planning plus a learned long-horizon value. The mechanism this benchmark isolates is not particular to greenhouses or to MPC. A short-horizon economic controller is myopic because the payoff lies beyond its horizon; supplying a *learned long-horizon value* restores far-sighted behaviour. This is the same structure that recurs across sequential decision-making: a shallow tree search made strong by a learned value network at its leaves in board games [34], a short-horizon model-predictive controller made strong by a learned terminal value in robotics [35, 36], and, unifying both, the approximate-dynamic-programming view of predictive control as rollout with a terminal cost-to-go approximation [37]. Our arm 4 is the greenhouse instance of that pattern, and the -3.1-to-$+3.6$ EUR m^{-2} swing in mean net value from pure NMPC to the RL-in-MPC hybrid is a direct measurement of what the learned terminal value is worth.

Beyond greenhouses. Building climate control and distributed-energy-resource scheduling (heat pumps, batteries, and storage operated against a time-varying tariff) share the greenhouse's struc-

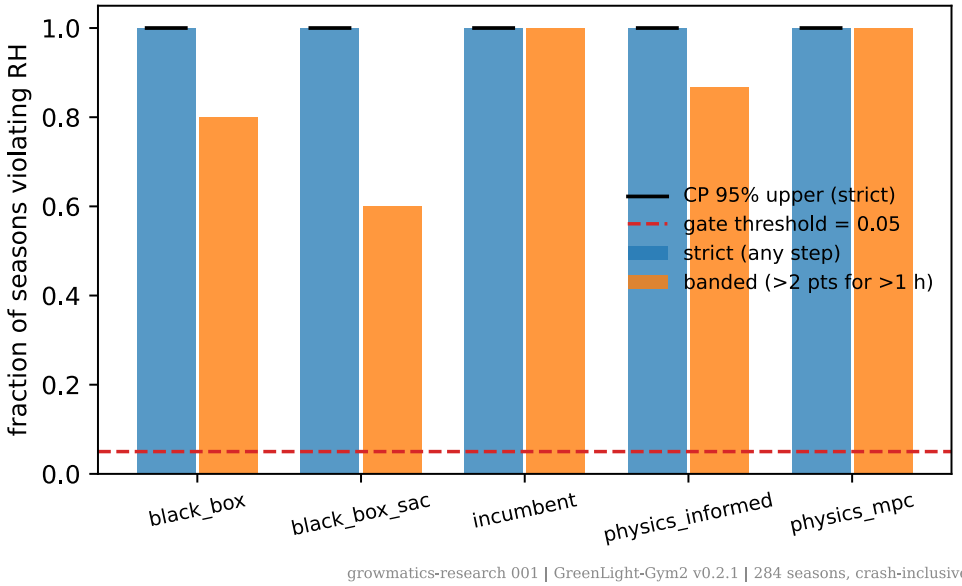


Figure 7: Violating-season rates against the pre-registered $\varepsilon = 5\%$ gate (crash-inclusive Clopper–Pearson uppers). The native economic objective saturates the gate for every arm.

Table 7: Paired arm-versus-arm net value (Wilcoxon signed-rank, matched conditions, Holm-corrected). n is matched (year $\times$ seed [$\times$ mismatch]) conditions; the deterministic incumbent's four conditions are underpowered ($p \geq 0.125$ is the floor at $n = 4$).

A	B	n	median diff	Cliff's δ	p (Holm)
RL-in-MPC	black-box PPO	20	+0.08	1.0	9.5×10^{-6}
RL-in-MPC	physics NMPC	120	+7.91	1.0	1.2×10^{-20}
RL-in-MPC	incumbent	4	+6.30	1.0	0.375
black-box PPO	physics NMPC	20	+6.60	1.0	9.5×10^{-6}
black-box PPO	incumbent	4	+6.24	1.0	0.375
physics NMPC	incumbent	4	−0.85	0.0	1.000

ture: a slow storage state, a hard operating limit, and an energy-dominated objective, a domain where learned control has already been reported to outperform a rule-based baseline on a heat-pump storage system [38]. The measurement transfers; the numbers do not. Solver reliability and a hard feasibility gate belong alongside economic value as deployment axes on any such system. The crossover budget and the size of the physics head start are testbed-specific and must be re-measured per domain.

6 Limitations

The study uses a single simulator and a single crop. The plant is deterministic: bit-reproducible within a platform, with floating-point tolerance across platforms. Wall-clock online cost is hardware- and load-stamped and valid only as a relative axis. The incumbent contributes only four independent seasons on the deterministic plant, so its gate row is flagged underpowered. Mismatch is a parametric perturbation of the nominal model; it preserves the model's functional form and is invis-

ible to the model-free arms, so it is an upper bound on the structured model error real greenhouses present. The rule-based reference is untuned and loss-making over this window; it anchors the Pareto reference, so "improvement over the incumbent" is read against a reference rather than a competitive baseline. The humidity metric, degree-hours above the bound, is not disease-grounded and does not separate a benign daytime excursion from a dangerous night-time near-saturation. The black-box corner is represented by PPO and, after equal-procedure random-search tuning, SAC (2M five-seed held-out net +3.21, positive by ∼100k steps with no mid-training valley, versus PPO's +3.50). Both learned arms make the data-efficiency claim, so it is not a PPO artifact; RecurrentPPO and other classes remain untested. The safety profile is analysed for the humidity class only: the run records also hold the temperature, CO_2, and lamp classes, and under the temperature band the physics-only NMPC accumulates roughly ten times the degree-hours of every other arm while the three learned arms, unlike the incumbent, exceed the band for over an hour in every season, a profile this paper leaves unanalysed.

Acknowledgments

This work was funded by Growmatics B.V. as part of its internal research program. The author thanks colleagues at Growmatics and Breda University of Applied Sciences for helpful discussions, and the authors of GreenLight-Gym2 and of the GreenLight model for releasing their simulators openly; weather data are from the Open-Meteo archive.

Reproducibility statement

Every number in this paper traces to a committed JSON run record (git commit, hardware stamp, grid, failures) in the released repository; the analysis pipeline regenerates all tables and figures from those records with one command. The plant is bit-deterministic within a platform (verified $\Delta = 0$ on repeat runs of every arm), and the NMPC seasons reproduce exactly only with the math libraries pinned to one thread (`OMP_NUM_THREADS=1`, as in the recorded run): under default threading the IPOPT path diverges over the 5760 solves and a season's net value shifts by about 0.1 EUR m^{-2}; PPO training follows the testbed's published recipe read from its pinned source. The source data underlying every figure and statistical test is the committed per-season run record; a per-artifact command map (which script and input file regenerates each table and figure) is in `manuscript/REPRODUCE.md`, and the anchor is `benchmark_r1-union.json`. The tables and figures regenerate via `examples/analyze.py` and all significance tests via `examples/stats.py`. The substrate is GreenLight-Gym2 v0.2.1 (commit `8bb5b28`); the controllers use CasADi 3.6.7 (which bundles IPOPT and the SUNDIALS/CVODES integrator), Stable-Baselines3 2.1.0 with sb3-contrib 2.1.0, Gymnasium 0.29.1, PyTorch 2.13.0, and NumPy 1.26.4.

Data and code availability

The benchmark code for all five arms, the controller implementations, the committed run records, and the analysis pipeline that regenerates every reported table and figure will be released as a frozen open-source repository at https://github.com/growmatics/greenhouse-control-benchmark (release v1.0.0). The maintained successor of that code, Phytotron [39], is public (version 0.1.0, AGPL-3.0-or-later). Its validation page documents that it reproduces this study's rule-based incumbent on all four held-out years and its physics-only NMPC season to six decimals. It ships no trained policy and no RL-in-MPC arm, and it gates seasons with crop-literature rules fixed after

this study's run, so the frozen repository remains the reference for the learned and hybrid arms and for the pre-registered gate. All inputs are public: weather data from Open-Meteo and the open GreenLight-Gym2 simulator (v0.2.1, pinned); no commercial-grower or otherwise proprietary data is used.

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